

Ivan Petrov

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Professional Summary

- Embedded Systems Engineer with 5 years of experience developing firmware for IoT devices and industrial controllers.
- Skilled in C, C++, RTOS concepts, microcontroller programming, communication protocols, and hardware debugging.
- Experienced in building reliable low-level software for resource-constrained systems.

Skills

Languages:

C, C++, Python

Embedded:

Firmware Development, RTOS, Microcontrollers, Device Drivers

Protocols:

UART, SPI, I2C, CAN, MQTT

Hardware:

Oscilloscope, Logic Analyzer, PCB Bring-up

Tools:

Keil, STM32CubeIDE, Git, CMake

Work Experience

Embedded Systems Engineer at VoltEdge Automation

Sofia, Bulgaria | Aug 2021 - Present

- Developed firmware for industrial sensor modules using C and STM32 microcontrollers.
- Implemented CAN and MQTT communication features for remote telemetry reporting.
- Reduced device boot time by optimizing initialization routines and peripheral checks.

Firmware Developer at Balkan IoT Labs

Plovdiv, Bulgaria | Jul 2019 - Jul 2021

- Built embedded software for smart metering and environmental monitoring devices.
- Created hardware diagnostics tools using Python scripts and serial communication.
- Supported board bring-up, firmware flashing, and field issue investigation.

Projects

Smart Sensor Gateway

Tech Stack: C, STM32, FreeRTOS, MQTT

Built firmware for a sensor gateway with task scheduling, telemetry publishing, and fault recovery.

CAN Diagnostics Tool

Tech Stack: C++, CAN Bus, Python, Qt

Created a desktop-supported diagnostic workflow for reading device status over CAN networks.

Education

B.Eng. in Electronics and Computer Engineering - Technical University of Sofia

Sofia, Bulgaria | 2015 - 2019

Certifications

- Embedded C Programming Certification
- ARM Cortex-M Microcontroller Development Certification